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Advanced RAG with Vector Databases and Retrievers

Today's goals
Complete any 3 learning items · 3/3
Watch 2 videos · 2/2
Complete a graded assessment

Today's Skill Points 134 XP
See skill progress

Module 1
Advanced Retrievers for RAG

Welcome to the Course
✓ Course Introduction
Video • 4 min
✓ Course Overview
Reading • 5 min
✓ RAG and Agentic AI Professional Certificate Overview
Video • 5 min
✓ Reading: Helpful Tips for Course Completion
Ungraded Plugin • 5 min

Work with Advanced Retrievers in LangChain
✓ Explore Advanced Retrievers in Langchain: Part 1
Video • 2 min
✓ Explore Advanced Retrievers in Langchain - Part 2
Video • 5 min
○ Lab: Build a Smarter Search with LangChain Context Retrieval
Ungraded App Item • 1h
✓ [Optional] Interactive LangChain Lesson Recap Podcast (AI-Powered)
Ungraded App Item • 10 min
✓ Practice Quiz: Work with Advanced Retrievers in LangChain
Practice Assignment • Grade: 85.71%

Work with Advanced Retrievers in LlamaIndex
✓ Advanced Retrievers in LlamaIndex
Video • 8 min
○ Lab: Explore Advanced Retrievers in LlamaIndex
Ungraded App Item • 1h
✓ [Optional] Interactive LlamaIndex Lesson Recap Podcast (AI-Powered)
Ungraded App Item • 10 min
✓ Practice Quiz: Work with Advanced Retrievers in LlamaIndex
Practice Assignment • Grade: 100%

Module Summary and Evaluation
○ **Summary and Highlights: Advanced Retrievers for RAG**
Reading • 3 min
○ Reading: Cheat Sheet: Advanced Retrievers for RAG
Ungraded Plugin • 15 min
○ Graded Quiz: Advanced Retrievers for RAG
Graded Assignment • 21 min
○ [Optional] Meet and Greet
Discussion Prompt • 2 min

Module 2
Build a Comprehensive RAG Application

Summary and Highlights: Advanced Retrievers for RAG

6 XP • Advanced RAG System Implementation

2:00 / 2:54 1x

Congratulations! You have completed this lesson. At this point in the course, you know:

- A LangChain retriever is an interface that returns documents based on an unstructured query
- There are several different types of LangChain retrievers
- The vector store-based retriever retrieves documents from a vector database
- A vector store-based retriever can be created directly from the vector store object with the retriever method by using similarity search or MMR
- That similarity search is when the retriever accepts a query and retrieves the most similar data
- MMR is a technique used to balance the relevance and diversity of retrieved results
- The multi-query retriever uses an LLM to create different versions of the query, generating a richer set of retrieved documents
- The self query retriever converts the query into two components, a string to look up semantically, and a metadata filter to accompany it
- The parent document retriever has two text splitters: a parent splitter that splits the text into large chunks to be retrieved, and a child splitter that splits the document into small chunks to generate meaningful embeddings
- The core LlamaIndex index types are the VectorStoreIndex, the DocumentSummaryIndex, and the KeywordTableIndex
- The VectorStoreIndex stores vector embeddings for each document chunk, is best suited for semantic retrieval, and is commonly used in pipelines that involve large language models
- The DocumentSummaryIndex generates and stores summaries of documents, which are used to filter documents before retrieving the full content, and is useful when working with large and diverse document sets
- The KeywordTableIndex extracts keywords from documents and maps them to specific content chunks, and is useful in hybrid or rule-based search scenarios
- The Vector Index Retriever uses vector embeddings to find semantically related content, and is ideal for general-purpose search and RAG pipelines
- The BM25 Retriever is a keyword-based method for ranking documents, and it retrieves content based on exact keyword matches rather than semantic similarity
- The Document Summary Index Retriever uses document summaries instead of the actual documents to find relevant content
- There are two versions of the Document Summary Index Retriever, one uses LLM, and the other uses semantic similarity
- The Auto Merging Retriever preserves context in long documents using a hierarchical structure, and uses hierarchical chunking to break documents into parent and child nodes
- The Recursive Retriever follows relationships between nodes and uses references such as citations in academic papers or metadata links
- The Query Fusion Retriever combines results from different retrievers using fusion strategies
- The fusion strategies supported by the Query Fusion Retriever are Reciprocal Rank Fusion, Relative Score Fusion, and Distribution-Based Fusion

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